Qian Lou

https://www.qlou.org/agentic-ai/ | qian.lou@ucf.edu | 812-558-6704

Research Profile

Qian Lou is a researcher and professor at the University of Central Florida, serving as Principal Investigator on NSF-, NIH-, and Amazon-supported projects. His research connects capable and efficient models, collaborative intelligence, and trustworthy collaboration to expand what AI agents can accomplish together. He develops model architectures, routing and orchestration methods, and safeguards for agent interactions and computation. His NIH-supported research studies knowledge-grounded collaborative agents for healthcare decision support. He received the **NSF CAREER Award in 2026** and previously worked as a Senior Research Scientist at Samsung AI Center.

Working Experience

University of Central Florida — Orlando, FL
Assistant Professor of Computer Science — September 2022 - Present

Samsung AI Center — Mountain View, CA
Senior Research Scientist — March 2021 - August 2022

Samsung AI Center — Mountain View, CA
Research Intern — May 2020 - August 2020

Education

Indiana University Bloomington
Ph.D., Computer Engineering — 2017 - 2021

Indiana University Bloomington
M.S., Computer Engineering — 2017 - 2019

Shandong University
B.S., Computer Science — 2013 - 2017

Awards

2026 NSF CAREER Award.
2026 ASEM-FL Rising Star Award.
2026 Amazon Research Award.
2025 Distinguished Speaker, Warren B. Nelms Annual AIoT Conference.
2022 Best Paper Award, Samsung AI Center.
2021 Luddy Outstanding Research Award, Indiana University Bloomington.
2020 Best Paper Finalist, ACM Parallel Architectures and Compilation Techniques (PACT).
2018 Best Paper Finalist, International Conference on Computer-Aided Design (ICCAD).

Grants and Research Support

2026 **NSF CAREER. Principal Investigator; OAC-2544047; $600K.**
Scalable, General-Purpose, and Automated Secure Computing for AI Infrastructure

2024 **NSF CISE Core / CSR Small. Principal Investigator; CNS-2413232; $599,725.**
System Support for Fast, Scalable, and Verifiable Fully Homomorphic Encryption

2025 **NSF SaTC. Principal Investigator; CCF-2523407; $124,786 UCF award.**
Collaborative Research: Privacy-Preserving Synthetic Data Generation

2026 **NIH/NIAID R01. Principal Investigator; $1.2M project over four years.**
Collaborative AI Agents for Bacterial Infections: A Knowledge-Grounded Framework to Support Clinical Practice Guidelines and Clinical Decision-Making

2026 **Amazon Research Award. Principal Investigator; $250K in AWS credits.**
Trainium-native controller-guided compression and alignment-preserving post-training

2026 **NSF CyberAI Scholarship for Service (SFS). Co-Principal Investigator; 2622846; $1,955,351 total award.**
CyberAI SFS: Educating Florida's Federal CyberAI Workforce for Secure AI in Space, Defense, and Modeling & Simulation

2026 **NSF Research Experiences for Undergraduates (REU).**
Support for undergraduate research experiences

2025 **Department of Defense Pathfinder Program. Principal Investigator; approximately $100K.**
LLM Security Testing

2024 **Arctic Wolf Networks, Inc. Co-Principal Investigator; approximately $160K.**
Industry-sponsored research on AI and cybersecurity

Open-Source and Artifacts

Model and reasoning-budget selection: R2-Router.
Efficient and trustworthy machine learning: UCF-ML-Research.
Programming agents with external checks: FHE-Coder benchmark and project resources.
Privacy-preserving computing and AI: UCF-Lou-Lab-PET.

Selected Research Contributions and Impact

- **Capable and efficient models.** DictFormer and weighted low-rank compression (ICLR 2022) make model capabilities smaller and more reusable, providing foundations for specialized agents. DictFormer was patented and translated into Samsung products.
- **Collaborative intelligence.** R2-Router (ICML 2026) chooses models and output budgets to balance answer quality and cost. Gated-memory routing (EMNLP 2026) and latency-aware orchestration (NeurIPS 2026) study what agents retain and how they coordinate. INFRAMIND and AgentServeSim (preprints) connect collaboration to serving conditions and simulation-based policy search.
- **Trustworthy collaboration.** MemIncept (ICML 2026) and conjunctive prompt attacks (ACL 2026) expose risks in agent memory and interaction. Clinical-text studies (EMNLP 2025/2026) examine factuality and watermarking. SecureRoute, DataSeal, and FHE-Coder address private computation and external checks of supported results and programs.
- **Healthcare as an integrating application (ongoing).** The NIH/NIAID R01 studies knowledge-grounded collaborative agents for bacterial infection decision support. The goal is to connect specialized capabilities, evidence coordination, and reliability evaluation to clinical practice guidelines. Clinical benefits remain to be validated, with clinician oversight essential.

Patents

[5] Wenbo Li, Zhipeng Mo, Yi Wei, Burak Uzkent, Qian Lou, Yilin Shen, Hongxia Jin, Fusion techniques for combining most significant bits and least significant bits of image data in image processing or other applications (US20240080423A1).

[4] Yen-chang Hsu, Ting Hua, Feixuan Wang, Qian Lou, Yilin Shen, Hongxia Jin, Machine learning model compression using weighted low-rank factorization (US20230106213A1).

[3] Qian Lou, Yen-Chang Hsu, Burak Uzkent, Ting Hua, Yilin Shen, Hongxia Jin, Small and fast transformer model for multi-modal or other tasks (US20230177338A1).

[2] Qian Lou, Yilin Shen, Hongxia Jin, Ting Hua, Yen-Chang Hsu, Small and fast transformer with shared dictionary (US20230104491A1).

[1] Qian Lou, Yilin Shen, Hongxia Jin, Method and system for secure, accurate and fast neural network inference (US20220121947A1).

Selected Agentic AI Publications

** Students and staff under my supervision. Original publication numbers are retained.*

[70] Jiaqi Xue*, Mengxin Zheng, Heng Huang, Qian Lou, Chain-of-Route: State-Aware LLM Routing for Multi-Turn Conversations. In Advances in Neural Information Processing Systems **(NeurIPS)**, 2026.

[69] Xi Shi*, Mengxin Zheng, Qian Lou, Learning Latency-Aware Orchestration for Multi-Agent Systems. In Advances in Neural Information Processing Systems **(NeurIPS)**, 2026.

[65] Rakibul Hasan Rajib*, Mengxin Zheng, Qian Lou, Learning What to Retain: Gated-Memory Routing for Efficient Collaboration in Multi-Agent LLM Systems. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)**, Main Conference, 2026.

[64] Rochana Prih Hastuti*, Rian Adam Rajagede, Mengxin Zheng, Qian Lou, Evaluating LLM Watermarking for Clinical EHR: Detectability, Style Shifts, and Helpfulness. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)**, Main Conference, 2026.

[62] Nan Yan, Qian Lou, and Jiarong Xing, MemIncept: Steering LLM Agents via Cooperative Stealthy Memory Injections. In International Conference on Machine Learning **(ICML)** 2026.

[61] Jiaqi Xue*, Qian Lou, Jiarong Xing, and Heng Huang, R2-Router: A New Paradigm for LLM Routing with Reasoning. In International Conference on Machine Learning **(ICML)** 2026.

[59] Nokimul Hasan Arif*, Qian Lou, and Mengxin Zheng, Conjunctive Prompt Attacks in Multi-Agent LLM Systems. In Annual Meeting of the Association for Computational Linguistics **(ACL)** 2026. Main Conference Oral.

[56] Yukuan Zhang*, Mengxin Zheng, and Qian Lou, SecureRoute: Input-Adaptive Encrypted Routing for Efficient and Accurate MPC-Based Transformer Inference. In Design Automation Conference **(DAC)** 2026.

[51] Mayank Kumar*, Jiaqi Xue, Mengxin Zheng, and Qian Lou, FHE-Coder: Secure Agentic Code Generation for Fully Homomorphic Encryption. In International Conference on Learning Representations **(ICLR)** 2026.

[46] Rochana Prih Hastuti*, Rian Adam Rajagede*, Mansour Al Ghanim*, Mengxin Zheng, Qian Lou, Factuality Beyond Coherence: Evaluating LLM Watermarking Methods for Medical Texts. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)** 2025.

[40] Muhammad Santriaji*, Jiaqi Xue*, Qian Lou, Yan Solihin, DataSeal: Ensuring the Verifiability of Private Computation on Encrypted Data. In IEEE Symposium on Security and Privacy **(S&P)** 2025 (Oakland). Qian Lou is the corresponding author.

[19] Yen-Chang Hsu, Ting Hua, Sungen Chang, Qian Lou, Yilin Shen, and Hongxia Jin. " Language model compression with weighted low-rank factorization ". In International Conference on Learning Representations **(ICLR)** 2022.

[18] Qian Lou, Ting Hua, Yen-Chang Hsu, Yilin Shen, and Hongxia Jin. "DictFormer: Tiny Transformer with Shared Dictionary". In International Conference on Learning Representations **(ICLR)** 2022.

Additional Publications

** Students and staff under my supervision. Original publication numbers are retained.*

[68] Yifei Zhao*, Qian Lou, Mengxin Zheng, RealM: A Unified Red-Teaming Benchmark for Physical-World VLMs. In Advances in Neural Information Processing Systems **(NeurIPS)**, Evaluations and Datasets Track, 2026.

[67] Jiaqi Xue*, Yifei Zhao*, Mansour Al Ghanim, Shangqian Gao, Ruimin Sun, Qian Lou, Mengxin Zheng, PRO: Enabling Precise and Robust Text Watermark for Open-Source LLMs. In Advances in Neural Information Processing Systems **(NeurIPS)**, 2026.

[66] Hedong Zhang*, Mengxin Zheng, Qian Lou, CipherFlow: Hardware-Aware Compiler Framework for Low-Latency Hybrid Secure Inference. In Advances in Neural Information Processing Systems **(NeurIPS)**, 2026.

[63] Mayank Kumar*, Qian Lou, Paulo Barreto, Martine De Cock, and Sikha Pentyala, FHAIM: Fully Homomorphic AIM for Private Synthetic Data Generation. In International Conference on Machine Learning **(ICML)** 2026.

[60] Ruizhi Zhu, Yanan Guo, Huize Li, Weidong Cao, Qian Lou, and Xin Xin, HBM-CASO: A Coordinated Approach to HBM System-Level and On-Die ECC. In ACM Annual International Symposium on Computer Architecture **(ISCA)** 2026.

[58] Erwin Eko Wahyudi*, Yan Solihin, and Qian Lou, Efficient Arithmetic-and-Comparison Homomorphic Encryption with Space Switching. In IEEE Symposium on Security and Privacy **(S&P)** 2026.

[57] Ahasan Kabir*, Jiaqi Xue, Mengxin Zheng, and Qian Lou, HW-Router: Hardware-Aware Routing for Scalable Multi-LLM Serving. In Design Automation Conference **(DAC)** 2026.

[55] Yifei Zhao*, Qian Lou, and Mengxin Zheng, SIF: Semantically In-Distribution Fingerprints for Large Vision-Language Models. In IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)** 2026.

[54] Jiaqi Xue*, Xin Xin, Wei Zhang, Mengxin Zheng, Qianqian Song, Minxuan Zhou, Yushun Dong, Dongjie Wang, Xun Chen, Jiafeng Xie, Liqiang Wang, David Mohaisen, Hongyi Wu, and Qian Lou, SoK: Can Fully Homomorphic Encryption Support General AI Computation? A Functional and Cost Analysis. In Proceedings on Privacy Enhancing Technologies **(PoPETs)** 2026.

[53] Fan Li, Mayank Kumar*, Ruizhi Zhu, Mengxin Zheng, Qian Lou, and Xin Xin, ReliaFHE: Resilient Design for Fully Homomorphic Encryption Accelerators. In ACM International Conference on Architectural Support for Programming Languages and Operating Systems **(ASPLOS)** 2026, pp. 1605-1621.

[52] Jiaqi Xue*, Mengxin Zheng, and Qian Lou, RobPI: Robust Private Inference against Malicious Client. In IEEE Conference on Secure and Trustworthy Machine Learning **(SaTML)** 2026.

[50] Chi Zhang*, Mengxin Zheng, Qian Lou, Hui Min Leung, and Fan Chen, VQEzy: An Open-Source Dataset for Parameter Initialization in Variational Quantum Eigensolvers. In IEEE International Conference on Acoustics, Speech, and Signal Processing **(ICASSP)** 2026.

[49] Chi Zhang*, Mengxin Zheng, Qian Lou, and Fan Chen, DiffQ: Unified Parameter Initialization for Variational Quantum Algorithms via Diffusion Models. In IEEE International Conference on Acoustics, Speech, and Signal Processing **(ICASSP)** 2026.

[48] Cheng Chu*, Qian Lou, Fan Chen, Lei Jiang, QNBAD: Quantum Noise-induced Backdoor Attacks against Zero Noise Extrapolation. In Network and Distributed System Security Symposium **(NDSS)** 2026.

[47] Jiaqi Xue*, Mayank Kumar*, Yuzhang Shang, Shangqian Gao, Rui Ning, Mengxin Zheng, Xiaoqian Jiang, Qian Lou, DictPFL: Efficient and Private Federated Learning on Encrypted Gradients. In Conference on Neural Information Processing Systems **(NeurIPS)** 2025.

[45] Mansour Al Ghanim*, Jiaqi Xue*, Rochana Prih Hastuti*, Mengxin Zheng, Yan Solihin, Qian Lou, Evaluating the Robustness and Accuracy of Text Watermarking Under Real-World Cross-Lingual Manipulations. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)** 2025.

[44] Yancheng Zhang*, Mengxin Zheng, Xun Chen, Jingtong Hu, Weidong Shi, Lei Ju, Yan Solihin, Qian Lou, zkVC: Fast Zero-Knowledge Proof for Private and Verifiable Computing. In Design Automation Conference **(DAC)** 2025.

[43] Yilan Zhu, Honghui You, Wei Zhang, Jiming Xu, Qian Lou, Shoumeng Yan, Lei Ju, DAHE: Parameter-adaptive and memory efficient FPGA acceleration of homomorphic encryption. In IEEE Transactions on Computers 2025.

[42] Yancheng Zhang*, Jiaqi Xue*, Mengxin Zheng, Mimi Xie, Mingzhe Zhang, Lei Jiang, Qian Lou, CipherPrune: Efficient and Scalable Private Transformer Inference. In International Conference on Learning Representations **(ICLR)** 2025.

[41] Rochana Prih Hastuti*, Rian Adam Rajagede, Mengxin Zheng, Qian Lou, Clinic-Prompt: Few-Shot Discrete Clinical Prompt Optimization. In Workshop on Large Language Models and Generative AI for Health at the AAAI Conference on Artificial Intelligence **(AAAI)** 2025.

[39] Yancheng Zhang*, Yuzhang Shang, Xun Chen, Mengxin Zheng, Qian Lou, Encrypted Data Pruning for Confidential Training of Deep Neural Networks. In Conference on Neural Information Processing Systems **(NeurIPS)** 2024.

[38] Jiaqi Xue*, Qian Lou, Mengxin Zheng, BadFair: Backdoored Fairness Attacks with Group-conditioned Triggers. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)** 2024.

[37] Mansour Al Ghanim*, Saleh Almohaimeed, Mengxin Zheng, Yan Solihin, Qian Lou, Jailbreaking LLMs with Arabic Transliteration and Arabizi. In Conference on Empirical Methods in Natural Language Processing **(EMNLP)** 2024.

[36] Deng, Xianglong, Shengyu Fan, Zhicheng Hu, Zhuoyu Tian, Zihao Yang, Jiangrui Yu, Dingyuan Cao, Dan Meng, Rui Hou, Meng Li, Qian Lou, Mingzhe Zhang, Trinity: A General Purpose FHE Accelerator. In IEEE/ACM International Symposium on Microarchitecture **(MICRO)** 2024.

[35] Yancheng Zhang*, Xun Chen, Qian Lou, HEBridge: Connecting Arithmetic and Logic Operations in FV-style HE Schemes. In ACM Conference on Computer and Communications Security **(CCS)** 2024 Workshop on Encrypted Computing & Applied Homomorphic Cryptography **(WAHC)**. (Theoretical and Foundational Work).

[34] Kunbei Cai, Zhenkai Zhang, Qian Lou, Fan Yao, WBP: Training-time Backdoor Attacks through Weight Bit Poisoning. In European Conference on Computer Vision **(ECCV)** 2024.

[33] Mengxin Zheng, Jiaqi Xue*, Zihao Wang, Xun Chen, Qian Lou, Lei Jiang, Xiaofeng Wang, SSL-Cleanse: Trojan detection and mitigation in self-supervised learning. In European Conference on Computer Vision **(ECCV)** 2024.

[32] Ardhi Wiratama Baskara Yudha*, Jiaqi Xue*, Qian Lou, Huiyang Zhou, Yan Solihin, BoostCom: Towards Efficient Universal Fully Homomorphic Encryption by Boosting the Word-wise Comparisons. In ACM International Conference on Parallel Architectures and Compilation Techniques **(PACT)** 2024.

[31] Qian Lou, Xin Liang*, Jiaqi Xue*, Yancheng Zhang*, Rui Xie, Mengxin Zheng, CR-UTP: Certified Robustness against Universal Text Perturbations on Large Language Models. In Annual Meeting of the Association for Computational Linguistics **(ACL)** 2024.

[30] Mengxin Zheng, Cheng Chu, Qian Lou, Nathan Youngblood, Mo Li, Sajjad Moazeni, Lei Jiang, OptoTFHE: An Electro-Optical Accelerator for Discretized TFHE. In IEEE/ACM International Symposium on Low Power Electronics and Design **(ISLPED)** 2024.

[29] Tho Nguyen, Ladda Thiamwong, Qian Lou, Rui Xie, Unveiling Fall Triggers in Older Adults: A Machine Learning Graphical Model Analysis. In Mathematics 2024.

[28] Mengxin Zheng, Jiaqi Xue*, Xun Chen, Yanshan Wang, Qian Lou, Lei Jiang, TrojFSP: Trojan Insertion in Few-shot Prompt Tuning. In Conference of the North American Chapter of the Association for Computational Linguistics **(NAACL)** 2024.

[27] Jiaqi Xue*, Mengxin Zheng, Ting Hua, Yilin Shen, Yepeng Liu, Ladislau Boloni, Qian Lou, TrojLLM: A Black-box Trojan Prompt Attack on Large Language Models. In Conference on Neural Information Processing Systems **(NeurIPS)** 2023.

[26] Mansour Al Ghanim*, Muhammad Santriaji*, Qian Lou, Yan Solihin, TrojBits: A Hardware Aware Inference-Time Attack on Transformer-Based Language Models. In European Conference on Artificial Intelligence **(ECAI)** 2023.

[25] Mengxin Zheng, Fan Chen, Lei Jiang, Qian Lou, PriML: An Electro-Optical Accelerator for Private Machine Learning on Encrypted Data. In International Symposium on Quality Electronic Design **(ISQED)** 2023.

[24] Qian Lou, Yepeng Liu*, Bo Feng, " TrojText: Invisible Test-time Trojan Insertion". In International Conference on Learning Representations **(ICLR)** 2023.

[23] Mengxin Zheng, Qian Lou, and Lei Jiang. " TrojViT: Trojan Insertion in Vision Transformers". In IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)** 2023.

[22] Mengxin Zheng, Qian Lou, Lei Jiang. "Primer: Privacy-Preserving Transformer on Encrypted Data". In Design Automation Conference **(DAC)** 2023.

[21] Qian Lou, Yen-Chang Hsu, Burak Uzkent, Ting Hua, Yilin Shen, and Hongxia Jin. " Lite-MDETR: A Lightweight Multi-Modal Detector ". In IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)** 2022.

[20] Lei Jiang, Qian Lou, Nrushad Joshi. "MATCHA: A Fast and Energy-Efficient Accelerator for Fully Homomorphic Encryption over the Torus". In Design Automation Conference **(DAC)** 2022.

[17] Mingqin Han, Yilian Zhu, Qian Lou, Zimeng Zhou, Shanqing Guo, Lei Ju. "coxHE: A software hardware co-design framework for FPGA acceleration of homomorphic computation". In Design, Automation and Test in Europe Conference **(DATE)** 2022.

[16] Bo Feng, Qian Lou, Geoffrey C. Fox, and Lei Jiang. "Low Latency Privacy-Preserving Text Analysis With GRU". In Conference on Empirical Methods in Natural Language Processing **(EMNLP)** 2021.

[15] Qian Lou, Lei Jiang "HEMET: A Homomorphic-Encryption-Friendly Privacy-Preserving Mobile Neural Network Architecture", In International Conference on Machine Learning **(ICML)** 2021

[14] Changsheng Zhao, Ting Hua, Yilin Shen, Qian Lou, and Hongxia Jin. "Automatic Mixed-Precision Quantization Search of BERT". In International Joint Conference on Artificial Intelligence **(IJCAI)** 2021.

[13] Qian Lou, Yilin Shen, Hongxia Jin, and Lei Jiang. "SAFENet: A Secure, Accurate, and Fast Neural Network Inference". In International Conference on Learning Representations **(ICLR)** 2021.

[12] Qian Lou, Wenjie Lu, Cheng Hong, and Lei Jiang. "Falcon: Fast Spectral Inference on Encrypted Data". In Conference on Neural Information Processing Systems **(NeurIPS)** 2020.

[11] Qian Lou, Bian Song, and Lei Jiang. "AutoPrivacy: Automated Layer-wise Parameter Selection for Secure Neural Network Inference". In Conference on Neural Information Processing Systems **(NeurIPS)** 2020.

[10] Qian Lou, Bo Feng, Geoffrey C. Fox, and Lei Jiang. "Glyph: Fast and Accurately Training Deep Neural Networks on Encrypted Data". In Conference on Neural Information Processing Systems **(NeurIPS)** 2020.

[9] Qian Lou, Sarath Janga, and Lei Jiang. "Helix: Algorithm/Architecture Co-design for Accelerating Nanopore Genome Base-calling." In ACM International Conference on Parallel Architectures and Compilation Techniques **(PACT)** 2020. Best Paper Nomination (4/197).

[8] Qian Lou, Feng Guo, Minje Kim, Lantao Liu, and Lei Jiang. "AutoQ: Automated Kernel-Wise Neural Network Quantization." In International Conference on Learning Representations **(ICLR)** 2020.

[7] Farzinah Zokaee, Qian Lou, N. Youngblood, Weichen Liu, and Lei Jiang. "LightBulb: A Photonic-Nonvolatile-Memory-based Accelerator for Binarized Convolutional Neural Networks." In Design, Automation and Test in Europe Conference **(DATE)** 2020.

[6] Qian Lou, Weichen Liu, Wenyang Liu, Feng Guo, and Lei Jiang, "MindReading: An Ultra-Low-Power Photonic Accelerator for EEG-based Human Intention Recognition," In Asia and South Pacific Design Automation Conference **(ASP-DAC)** 2020.

[5] Qian Lou and Lei Jiang. "SHE: A Fast and Accurate Deep Neural Network for Encrypted Data." In Conference on Neural Information Processing Systems **(NeurIPS)** 2019.

[4] Weichen Liu, Wenyang Liu, Yichen Ye, Qian Lou, Yiyuan Xie, and Lei Jiang. "HolyLight: A Nanophotonic Accelerator for Deep Learning in Data Centers." In Design, Automation and Test in Europe Conference **(DATE)** 2019.

[3] Qian Lou, Wujie Wen, and Lei Jiang. "3DICT: a reliable and QoS capable mobile process-in-memory architecture for lookup-based CNNs in 3D XPoint ReRAMs". In IEEE/ACM International Conference on Computer-Aided Design **(ICCAD)** 2018.

[2] Qian Lou, and Lei Jiang. "BRAWL: A Spintronics-Based Portable Basecalling-in-Memory Architecture for Nanopore Genome Sequencing". IEEE Computer Architecture Letters, 2018.

[1] Qian Lou, Mengying Zhao, Lei Ju, Chun Xue, Jingtong Hu, and Zhiping Jia. "Runtime and reconfiguration dual-aware placement for SRAM-NVM hybrid FPGAs." IEEE Non-Volatile Memory Systems and Applications Symposium **(NVMSA)** 2017.

Mentoring

Since joining UCF in Fall 2022, mentored 15+ students whose research has produced 60+ papers in leading AI, security, and systems venues. Graduated students include Mansour Al Ghanim (2025), faculty at King Khalid University, and Ardhi Wiratama Baskara Yudha, Senior Silicon Design Engineer at AMD.

Teaching

University of Central Florida: CDA 5106 Advanced Computer Architecture (Fall 2025, Fall 2023, Fall 2022); CAP 6614 Current Topics in Machine Learning (Spring 2025, Spring 2024, Spring 2023).

Indiana University Bloomington: IFT 4030/7030 Machine Learning for Signal Processing (Spring 2020); ENGR-E599 Introduction to Intelligent Systems (Fall 2019).

Invited Talks

2025 Distinguished Speaker, Warren B. Nelms Annual AIoT Conference at University of Florida.

2025 NSF CSR PI Meetings.

2024 Distinguished Speaker, Florida State University.

Professional Service

Workshop Chair: DAC 2026 SPGAI, Secure and Private System Design for Generative & Agentic AI.

NSF Panelist (2023-2025).

Area Chair: ACL, NeurIPS, ICLR.

Program Committee: CCS, NDSS, ISCA, DAC, ICCAD.

Track Chair: Hardware Security, GLSVLSI 2025-2026.

Senior/Program Committee: AAAI, CVPR, ACL, EMNLP, ICML, TMLR, PETs, TIFS.

Additional service: Session Chair, PACT 2024; Local Area Chair, ISCA 2023; UCF CS Graduate Committee Member; ESWEEK 2023 Design Contest Organizer.